

Evaluation of PCL/PEG Composites for Restorative Applications Within the Trachea

Cole R. Pearson¹, Solaleh Miar¹, Gregory R. Dion², Teja Guda¹

¹Department of Biomedical Engineering, University of Texas at San Antonio, San Antonio, TX

²US Army Institute of Surgical Research, JBSA Fort Sam Houston, TX 78234

Introduction: Our research aims to determine the ideal material for use in a mucoadhesive patch to treat tracheal wall injuries. A polycaprolactone (PCL)/polyethylene glycol (PEG) composite membranous patch is proposed to (1) attach to the mucous layer within the trachea in a sutureless manner, (2) enable degradation based controlled drug release, and to (3) be capable of production in large batches permitting intra-procedural sizing by a clinician to fit the injury site. In this preliminary research, our goal is to determine material suitability specifically in terms of being muco-adhesive, and then to benchmark the mechanical survival properties of that material.

Materials and Methods: PCL (Mn= 80,000, Sigma-Aldrich) was spun in random orientation by electrospinning (Fibers diameter: 4 μ m) and then different coatings were applied. Samples were either soaked in PEG, or flocked with PEG to create different flock densities (1x, 2x, 3x). Samples morphology was characterized by Scanning Electron Microscopy (SEM) and then stiffness, degradation rate, swelling behavior, and mucoadhesion properties were studied for all groups. Samples were placed in a PBS solution and remaining mass was measured over 14 days in order to evaluate the degradation rate. Swelling behavior of the patches were studied as the relative weight change of the wet and dry samples at different time points. Mucoadhesion testing was conducted on fresh porcine trachea tissue followed by testing in lap shear mode (UStretch, CellScale, Waterloo, ON). Quantitative data were compared using one-way Analysis of Variance (ANOVA) followed by Tukey's test for *post hoc* determination of significant differences at $p < 0.05$.

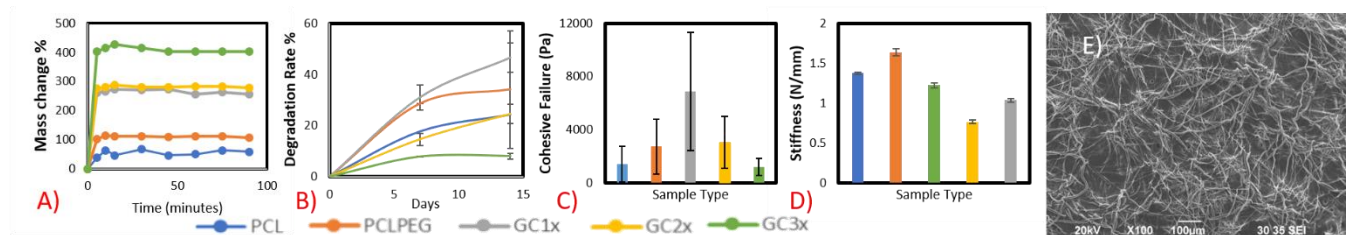


Figure 1: (A) Swelling behavior, (B) Degradation testing, (C) Mucoadhesion, (D) Stiffness measurement (E) Representative SEM

Results and Discussion: Random fiber structure and surface area was observed in the SEM images (Fig 1E). Results from the mucoadhesion test (Fig 1C) show that 1x flocked samples were the most mucoadhesive, with an observed cohesive failure stress of more than 6 kPa. The least mucoadhesive were the 3x flocked, although it was not significant compared to the uncoated PCL samples. Absorption testing showed that the 3x flocked group absorbed the most water, with uncoated PCL absorbing the least (Fig 1A). Degradation testing showed that 1x flocked samples degrade the most and the quickest, with 3x flocked degrading the slowest and the least (Fig 1B). PEG coated samples degraded as fast as 1x flocked initially, but degradation slowed after 7 days. No significant differences in mechanical properties were observed between groups (Fig 1D). Results showed a stiffness of 1.04 ± 0.02 N/mm, degradation of $31.0 \pm 4.8\%$ by 7 days and $46.5 \pm 5.7\%$ at 14 days, a water absorption peak of $273 \pm 8\%$, and an average mucoadhesive failure stress of 6.88 ± 4.44 kPa.

Conclusion: The most mucoadhesive samples of the composite material tested was the 1x flocked samples, which means that this material will likely be best suited for the intended use, as high muco-adhesion is the most important property of our desired material. Stiffness testing showed that the PEG coated were the stiffest, almost twice as stiff as the 2x flocked samples, which were the least stiff. According to the material design and characterization, PCL-PEG samples with 1x flock density are a suitable material for further *in vitro* and *in vivo* functional analyses.

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